Active Filters, Analog Noise Cancellation and High-Frequency Amplifiers

Circuit Theory Course Project – Corrected and Completed Report

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Semester	Fall 2025 this revision: October 2026
Tools	LTspice, ngspice 42, MATLAB/Simulink (Simscape), Python (NumPy, SciPy, SymPy)

Summary

This report replaces the handwritten solution pages of the original submission with typed derivations. Every result was checked again, the errors found are corrected, and the missing parts are added.

- **Part I – Sharif Analog Filter (SAF).** The transfer functions are derived from the node equations of the circuit as built. The design ($R_F = 8\,\text{k}\Omega$, $R_G = R_Q = 200\,\text{k}\Omega$) is confirmed: $f_c = 19.89\,\text{kHz}$ and $Q = 0.75$. The comparison table with percentage differences, which the original lacked, is added. The notch filter of the bonus question existed only as a screenshot; it is rebuilt as an LTspice file and simulated.
- **Part II – all-pass filters and noise cancellation.**
  - The 45 µs delay stage as built was not an all-pass filter. Its delay had been tuned by unbalancing a resistor, so its gain dropped by 2 dB near 10 kHz. The corrected values are $R_1 = R_2 = 1.125\,\text{k}\Omega$ and $R_3 = 1\,\text{k}\Omega$: they give a true all-pass response (flat within 0.04 dB) and the same 45 µs delay.
  - The delay is now verified at five test frequencies.
  - The complete chain reduces the noise in the project audio by about 33 dB.
- **Part III – high-frequency amplifiers.** All numerical answers of the original were correct, and are repeated here with the derivations typed. The two sets of plots that the assignment requires (Tasks 4.2 and 5.3) were missing; they are added. A more accurate common-emitter gain is given next to the value from the assignment's simplified formula.

Schematics, models, scripts and audio: https://github.com/seyedkiarash/seyedkiarash.github.io/tree/main/projects/circuit-filters-noise-cancellation.

Part I – Sharif Analog Filter

1 Transfer functions

The SAF is a state-variable (KHN) filter: amplifier A1 forms the high-pass output V_1 (HP), and two inverting integrators $R_{F1}C_1$ and $R_{F2}C_2$ produce the band-pass output V_2 (BP) and the low-pass output V_3 (LP). The non-inverting input of A1 sees the input through R_G, ground through R_Q and the band-pass output through R_4. The inverting input sees HP through R_2 and LP through R_1. With ideal op-amps ($v_+ = v_-$, no input current):

$$v_+ = \frac{V_i/R_G + V_2/R_4}{1/R_G + 1/R_Q + 1/R_4}, \qquad v_- = \frac{V_1/R_2 + V_3/R_1}{1/R_1 + 1/R_2},$$

$$V_2 = -\frac{V_1}{sR_{F1}C_1}, \qquad V_3 = -\frac{V_2}{sR_{F2}C_2}.$$

Solving the four equations (the algebra was also checked with SymPy) for the internal resistors $R_1 = R_2$ gives three transfer functions with a common denominator

$$D(s) = s^2 + \frac{\omega_n}{Q}s + \omega_n^2, \qquad \omega_n = \frac{1}{\sqrt{R_{F1}R_{F2}C_1C_2}}, \qquad Q = \frac{1}{2}\left(1 + \frac{R_4}{R_G \parallel R_Q}\right)\sqrt{\frac{R_{F1}C_1}{R_{F2}C_2}},$$

$$H_{HP}(s) = \frac{A\,s^2}{D(s)}, \qquad H_{BP}(s) = -\frac{R_4}{R_G}\,\frac{(\omega_n/Q)\,s}{D(s)}, \qquad H_{LP}(s) = \frac{A\,\omega_n^2}{D(s)}, \qquad A = \frac{2\,(R_4 \parallel R_Q)}{R_G + R_4 \parallel R_Q}.$$

The high-pass and low-pass outputs have the same pass-band gain A. The band-pass output is inverted, and its gain at ω_n is $-R_4/R_G$, independent of Q.

2 Design for $f_c = 20\,\text{kHz}$ and $Q = 0.75$

The assignment fixes $R_G = R_Q$ and $R_{F1} = R_{F2} = R_F$; the internal parts are $R_1 = R_2 = R_4 = 50\,\text{k}\Omega$ and $C_1 = C_2 = 1\,\text{nF}$.

- $f_c = 1/(2\pi R_F C) = 20\,\text{kHz}$ gives $R_F = 7.958\,\text{k}\Omega$. The nearest standard value, $8\,\text{k}\Omega$, gives $f_c = 19.89\,\text{kHz}$ (-0.5%).
- With $R_G = R_Q$, $R_G \parallel R_Q = R_G/2$, so $Q = \frac{1}{2}(1 + 2R_4/R_G) = 0.75$ gives $R_G = 4R_4 = 200\,\text{k}\Omega$.
- The resulting gains are $A = 1/3$ ($-9.54\,\text{dB}$) for HP and LP, and $-R_4/R_G = -0.25$ ($-12.04\,\text{dB}$) for BP at f_c.

These are the values the team used. The design is correct.

3 Simulation and comparison

The LTspice schematic was simulated in two ways. Converted to ngspice, it was run with ideal op-amps and with the OP747 modelled as a single-pole amplifier ($A_0 = 10^5$, GBW $= 1.8\,\text{MHz}$). The AC sweep runs from 1 to 100 kHz. f_c is the band-pass peak, and $Q = f_c/\text{BW}_{-3\,\text{dB}}$ is measured on the band-pass output.

Table 1. Analytical and simulated SAF parameters.

Quantity	Analytical	Simulated, ideal op-amp	Simulated, GBW 1.8 MHz	Difference (GBW)
f_c	19.894 kHz	19.894 kHz	20.110 kHz	+1.1%
Q	0.750	0.750	0.759	+1.2%
LP gain at DC	−9.54 dB	−9.54 dB	−9.54 dB	0
HP gain at 100 kHz	−9.54 dB (at ∞)	−9.51 dB	−9.17 dB	–
BP gain at f_c	−12.04 dB	−12.04 dB	−11.75 dB	+0.3 dB

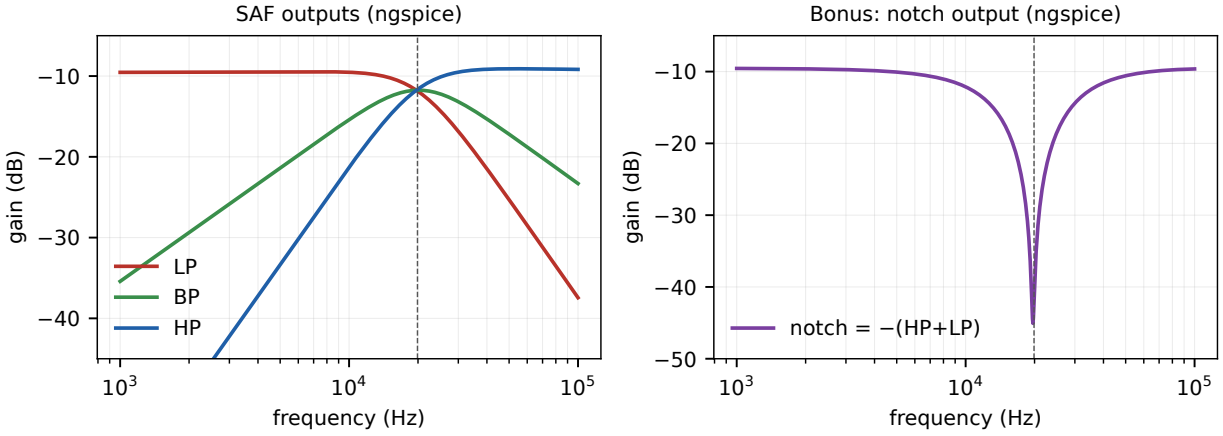


Figure 1. Left: SAF outputs (GBW model); the three curves cross at f_c, where $|H_{HP}| = |H_{LP}| = AQ$. Right: notch output of the bonus circuit.

With ideal op-amps the simulation reproduces the analysis exactly. The finite gain–bandwidth of the op-amp adds an extra phase lag in each integrator. This raises f_c and Q by about 1%, which is the main source of difference. Resistor tolerances act on Q through $R_4/(R_G \parallel R_Q)$: a 1% error in R_4, or in both R_G and R_Q, changes Q by 0.33%. The two integrators must match ($R_{F1}C_1 = R_{F2}C_2$) to keep the HP and LP gains equal. This is why integrated filters such as the UAF42, with laser-trimmed internal R_1, R_2, R_4, C_1, C_2, are more stable than discrete builds: the critical ratios are fixed on one die and drift together with temperature.

4 Bonus: notch filter

Because the HP and LP outputs have the same gain and sign, adding them cancels both numerator terms at ω_n:

$$H_N(s) = -(H_{HP}(s) + H_{LP}(s)) = -A\,\frac{s^2 + \omega_n^2}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}.$$

A fourth op-amp in an inverting summer ($R_3 = R_5 = R_6 = 10\,\text{k}\Omega$) adds the two outputs; this is the circuit in the team's screenshot. Its LTspice file was missing from the submission and is rebuilt as `saf_notch.asc`. With ideal op-amps the notch is infinitely deep at 19.89 kHz. With the GBW model it reaches −45 dB at 19.7 kHz, about 36 dB below the pass band (Fig. 1). The team's LTspice plot shows only −35 dB because its linear sweep has 99 Hz steps and misses the bottom of the notch.

Part II – All-Pass Filters and Analog Noise Cancellation

5 First-order all-pass filter

The input reaches the non-inverting input through an RC low-pass, $v_+ = V_{in}/(1 + sRC)$. Because the two R_1 resistors are equal, $V_{out} = 2v_+ - V_{in}$. Hence

$$H(s) = \frac{1 - sRC}{1 + sRC}, \qquad |H(j\omega)| = 1, \qquad \angle H(j\omega) = -2\arctan(\omega RC), \qquad \tau_g(\omega) = \frac{2RC}{1 + (\omega RC)^2}.$$

The pole and zero are mirror images, so the magnitude is 1 at every frequency. The phase falls from 0° at DC through −90° at $\omega = 1/RC$ to −180°. A sine passes with its amplitude unchanged and is delayed by τ_g; low frequencies are delayed by $2RC$. The product RC alone fixes the frequency of any given phase shift. For example, $\omega_{90^\circ} = 1/RC$; R_1 only has to be matched.

6 Second-order all-pass filter (the delay stage)

6.1 Transfer function of the circuit as built

In the team's circuit (Fig. 4 of the assignment), op-amp U1 with C_1, C_2, R_1 and R_2 forms a second-order section. U2 sums $-(R_{42}/R_3)\,v_{U1}$ and $-(R_{42}/R_{41})\,v_{in}$. Solving the node equations gives

$$H(s) = -\frac{R_{42}}{R_{41}} \cdot \frac{C_1C_2R_1R_2\,s^2 + [R_1(C_1 + C_2) - C_1R_2\,R_{41}/R_3]s + 1}{C_1C_2R_1R_2\,s^2 + R_1(C_1 + C_2)\,s + 1}.$$

The denominator gives $\omega_0 = 1/\sqrt{R_1R_2C_1C_2}$ and $Q = \sqrt{R_2/R_1}\cdot\sqrt{C_1C_2}/(C_1 + C_2)$. Unlike the first-order section, the second-order one has two parameters, ω_0 and Q, so the steepness of the phase transition can be set as well as its frequency. The response is all-pass only if the middle coefficient of the numerator is the negative of that of the denominator, and $R_{42} = R_{41}$ for unit gain. With $C_1 = C_2 = C$ and $R_1 = R_2 = R$:

$$\frac{R_{41}}{R_3} = 4 \quad\Longrightarrow\quad H(s) = -\left(\frac{1 - sRC}{1 + sRC}\right)^2, \qquad \angle H = 180^\circ - 4\arctan(\omega RC), \qquad \tau_g(0) = 4RC.$$

This equal-component section has a double real pole ($Q = 1/2$). It behaves like two first-order sections in series, with twice the phase shift and twice the delay. The leading minus sign is the phase inversion that the noise-cancelling chain uses.

6.2 Error in the original values and correction

The team built the stage with $R_1 = R_2 = 1.25\,\text{k}\Omega$, $C = 10\,\text{nF}$, $R_3 = 1.125\,\text{k}\Omega$ and $R_{41} = R_{42} = 4\,\text{k}\Omega$. Then $R_{41}/R_3 = 3.56 \neq 4$, so the numerator is not the mirror image of the denominator. The low-frequency delay came out at the required 45 µs ($\tau_g(0) = (R_{41}/R_3)\,RC = 3.56 \times 12.5\,\mu\text{s} = 44.4\,\mu\text{s}$) only because the

all-pass condition was broken. As a result, the gain falls by 2.1 dB near 10 kHz (Fig. 2). The correct design keeps the all-pass condition and sets the delay through RC:

$$4RC = 45\,\mu\text{s} \;\Rightarrow\; RC = 11.25\,\mu\text{s} \;\Rightarrow\; R_1 = R_2 = 1.125\,\text{k}\Omega\ (C = 10\,\text{nF}), \qquad R_3 = R_{41}/4 = 1\,\text{k}\Omega.$$

The two numbers 1.125 kΩ and 1 kΩ had evidently been placed in the wrong positions. The corrected schematic is `stage3_allpass_delay_45us_corrected.asc`. The same values are corrected in the Simulink model (`anc_system_corrected.slx`).

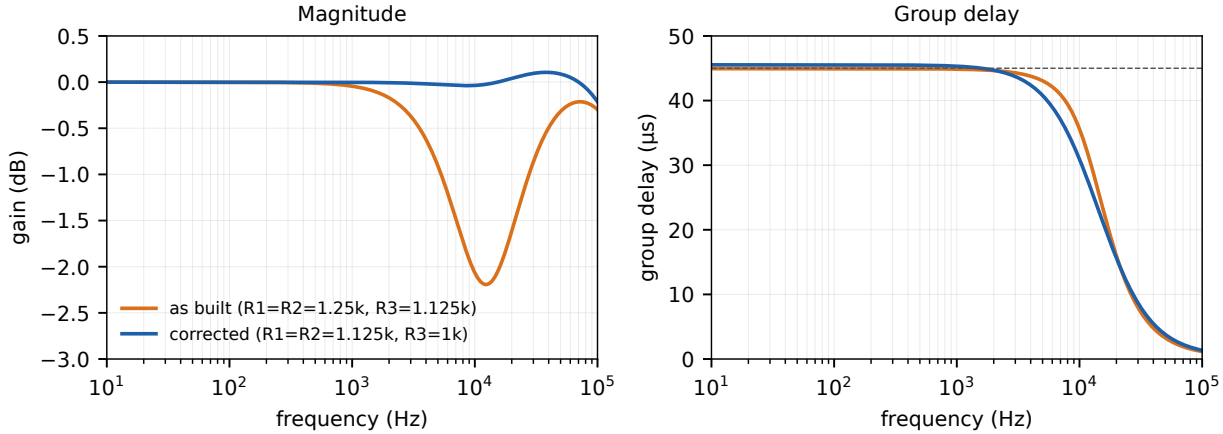


Figure 2. Delay stage, ngspice with the OP747 GBW model. As built, the magnitude is not flat; corrected, it stays within 0.04 dB up to 20 kHz, and the delay stays at 45 µs up to about 3 kHz.

6.3 Verification of the delay

Each stage was driven with a 1 V sine. The delay is the time between rising zero crossings of the input and the inverted output (Fig. 3), and the amplitude ratio is measured on the same waveforms.

Table 2. Transient test of the delay stage (ngspice, GBW model).

	As built		Corrected	
Frequency	Delay	Amplitude ratio	Delay	Amplitude ratio
200 Hz	44.9 µs	1.000	45.5 µs	1.000
500 Hz	44.9 µs	0.999	45.5 µs	1.000
1 kHz	44.9 µs	0.995	45.5 µs	1.000
2 kHz	44.8 µs	0.981	45.3 µs	0.999
5 kHz	44.3 µs	0.901	43.9 µs	0.997

With ideal op-amps the corrected stage gives exactly 45.0 µs at low frequency; the op-amp bandwidth adds 0.5 µs.

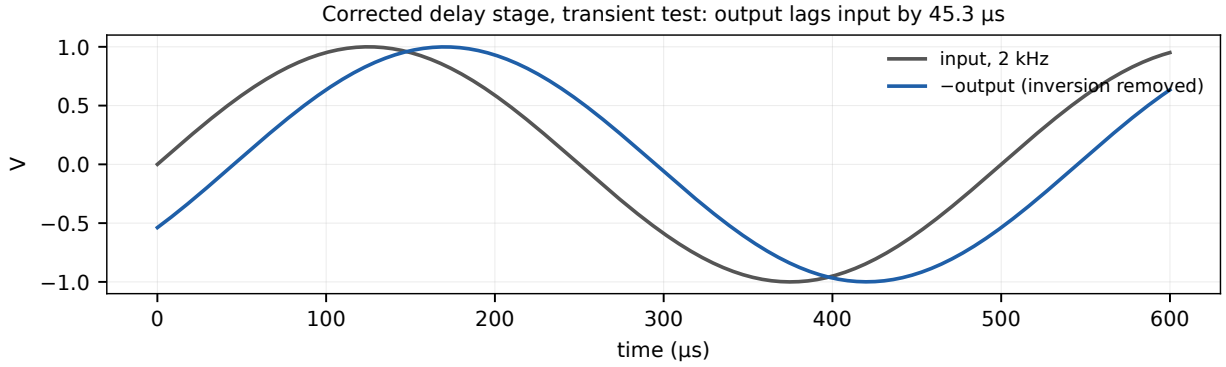


Figure 3. Corrected delay stage with a 2 kHz input.

7 Higher-order all-pass filters

Cascading N sections multiplies their responses. The magnitude stays 1, the phases add, and so do the group delays (a cascade of N first-order sections has $\tau_g(0) = 2NRC$). The impulse response of one first-order section is $h(t) = -\delta(t) + \frac{2}{RC}e^{-t/RC}u(t)$: an impulse followed by a single exponential. For a cascade the impulse response is the convolution of the sections' responses. Its energy moves to later times, near the group delay, and it acquires alternating lobes, because the phase now changes quickly with frequency (Fig. 4).

- **Order and duration.** The duration of the impulse response grows roughly in proportion to the order.
- **Phase.** The total phase shift grows by 180° for each first-order section (360° for each second-order one).
- **Visual difference.** A 10th-order response is a long, ringing burst centred near τ_g, whereas a 1st-order response is a single spike and decay.

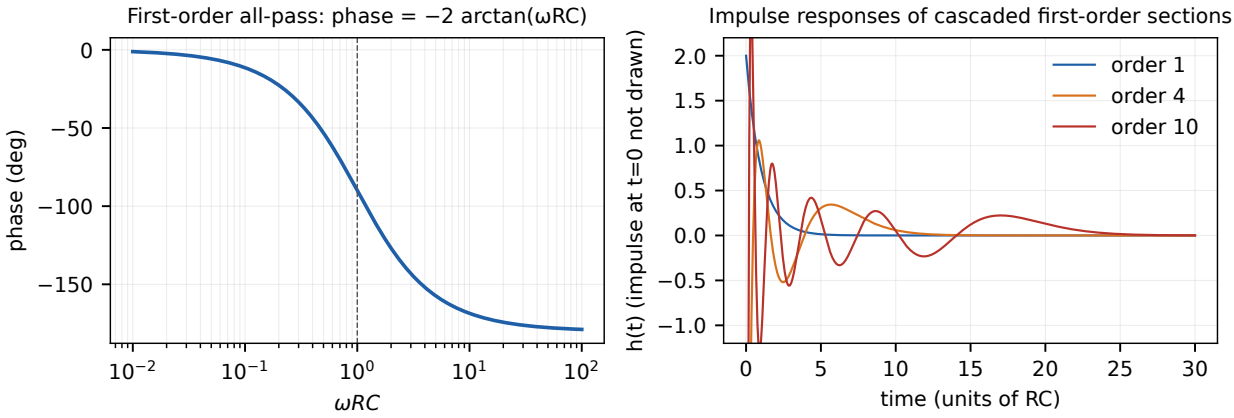


Figure 4. Left: phase of a first-order section. Right: impulse responses of cascades of 1, 4 and 10 equal sections (the initial $\pm\delta$ term is not drawn).

8 Analog noise cancellation

8.1 Why a delay is needed

The reference microphone picks up the noise before it reaches the listener. The acoustic path to the point of cancellation, and the electronics themselves, delay the noise that has to be cancelled. The anti-noise must arrive at the same time and with opposite sign. A phase error φ leaves a residual of $2|\sin(\varphi/2)|$ times the noise, so even $\varphi = 10°$ limits the reduction to about 15 dB. The delay stage therefore aligns the anti-noise in time with the noise it cancels.

8.2 The three stages

1. **Gain of 10.** A non-inverting amplifier with $R_f = 9\,\text{k}\Omega$ and $R_g = 1\,\text{k}\Omega$, so $1 + R_f/R_g = 10$. A 1 V test sine gives 10 V.
2. **Delay.** The corrected all-pass stage above: 45 µs, gain −1.
3. **Combination.** The assignment asks for a subtractor. Because the delay stage already inverts the noise, the team instead used an inverting summer (1 kΩ resistors) followed by an inverter, so the output is music + noise − delayed noise. This is equivalent to a subtractor. With test inputs of 1 V and 2 V the output is 3 V, as expected.

8.3 System results

The chain was built in Simulink (Simscape: band-limited op-amps with $A = 2 \times 10^5$ and resistors) and driven with the two recordings. The noise inside the recording is 10.06 times the isolated noise and lags it by about 30 µs; this was found by a least-squares fit, which also gives an estimate of the clean music.

- **Simulink output, original values:** noise reduced by 33.3 dB; signal-to-noise ratio from 8.4 dB to 41.6 dB (Fig. 5).
- **Corrected values:** a Python model of the same transfer functions gives 33.3 dB. The model reproduces

the Simulink output to within 0.5%. The correction leaves the result unchanged because the noise lies below 2 kHz, where both versions of the stage are nearly ideal.

- **Limiting factor:** the mismatch between the specified 45 µs and the 30 µs delay present in the recordings. With $RC = 7.5\,\mu\text{s}$ ($R_1 = R_2 = 750\,\Omega$) the same circuit would reduce the noise by about 44 dB.

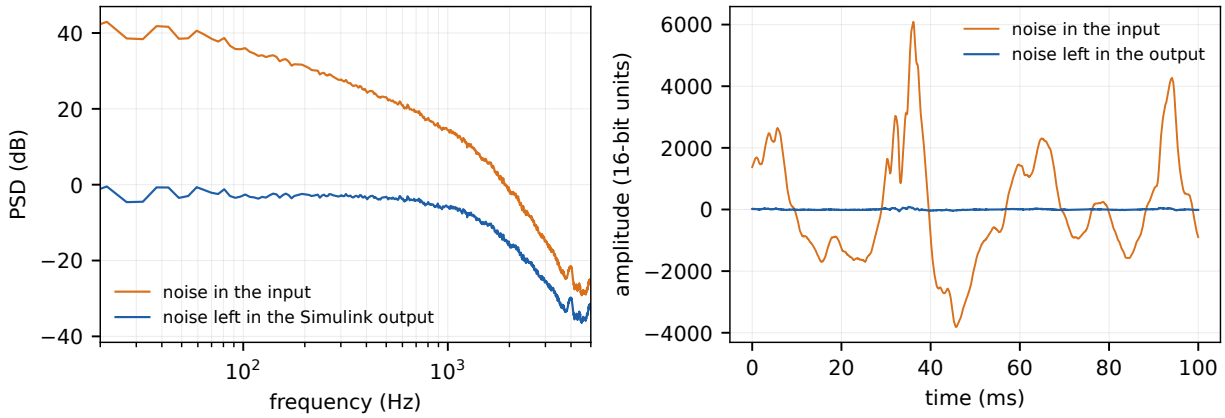


Figure 5. Noise before and after the cancellation chain (Simulink output).

Part III – High-Frequency Response of Transistor Amplifiers

9 Model parameters (Tasks 1 and 2)

Table 3. Small-signal parameters.

MOSFET		BJT	
$g_m = 2I_D/V_{OV}$	4.0 mA/V	$g_m = I_C/V_T$	80 mA/V
$r_o = V_A/I_D$	40 kΩ	$r_\pi = \beta_0/g_m$	1.5 kΩ
$C_{gs} = (\frac{2}{3}WL + WL_{ov})C_{ox}$	59.7 fF	$r_o = V_A/I_C$	37.5 kΩ
$C_{gd} = WL_{ov}C_{ox}$	6.4 fF	$C_\pi = \tau_F g_m + 2C_{je0}$	5.20 pF
		$C_\mu = C_{\mu 0}/(1 + V_{CB}/V_{0c})^m$	0.125 pF

The three expressions for g_m in Eq. (2) agree: $\mu_n C_{ox}(W/L)V_{OV} = 400\mu \cdot 40 \cdot 0.25 = 4\,\text{mA/V}$.

10 Unity-gain frequency (Tasks 3 and 4)

MOSFET. With the drain shorted, KCL at the drain gives $I_o = (g_m - sC_{gd})V_{gs}$, and at the gate $I_i = s(C_{gs} + C_{gd})V_{gs}$. Hence

$$A_i(s) = \frac{g_m - sC_{gd}}{s(C_{gs} + C_{gd})} \approx \frac{g_m}{s(C_{gs} + C_{gd})} \quad (g_m \gg \omega C_{gd}), \qquad f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} = 9.63\,\text{GHz}.$$

BJT. With the collector shorted, $I_b = (1/r_\pi + s(C_\pi + C_\mu))V_\pi$ and $I_c = (g_m - sC_\mu)V_\pi$. Hence

$$h_{fe}(s) = \frac{g_m - sC_\mu}{1/r_\pi + s(C_\pi + C_\mu)} \approx \frac{\beta_0}{1 + s/\omega_\beta}, \qquad \beta_0 = g_m r_\pi, \quad \omega_\beta = \frac{1}{r_\pi(C_\pi + C_\mu)}.$$

$|h_{fe}(j\omega_T)| = 1$ gives $\omega_T = \omega_\beta\sqrt{\beta_0^2 - 1} \approx \beta_0\omega_\beta = g_m/(C_\pi + C_\mu)$. The base resistance r_x does not affect h_{fe}, because the base is driven by a current source. Numerically, $f_\beta = 19.9\,\text{MHz}$ and $f_T = 2.39\,\text{GHz} = \beta_0 f_\beta$. The exact expressions, including the right-half-plane zeros, cross 0 dB at 9.67 GHz and 2.39 GHz.

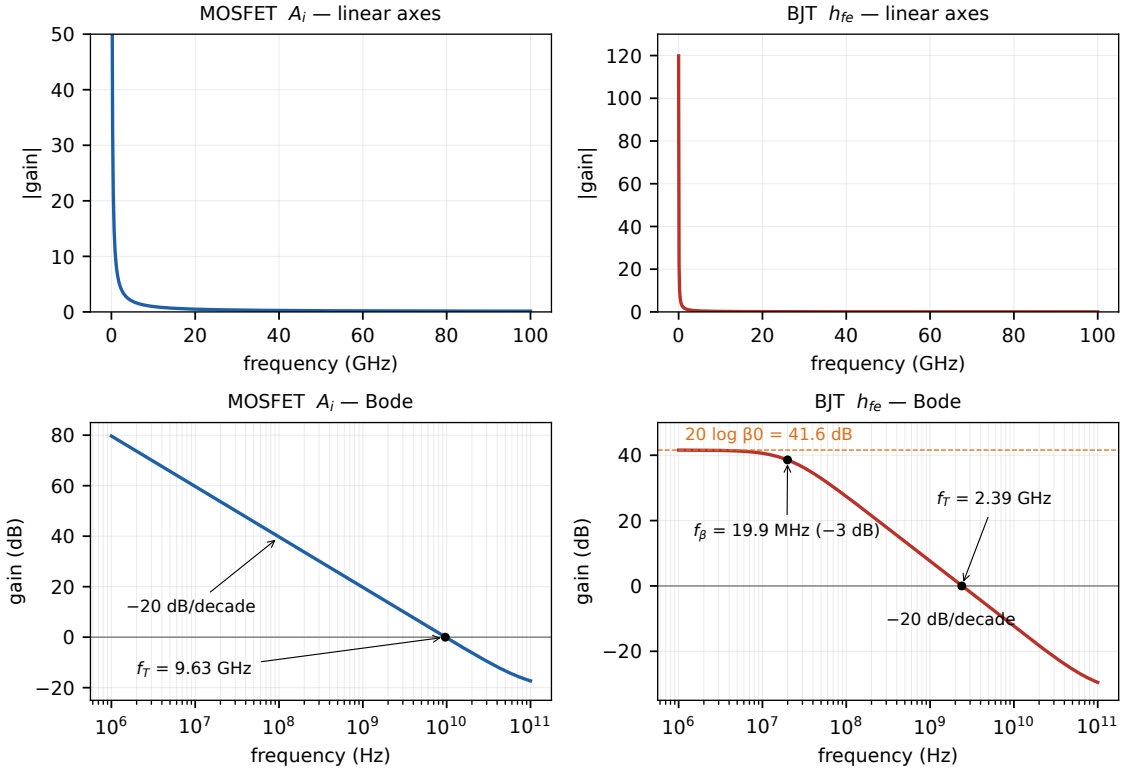


Figure 6. Short-circuit current gain from 1 MHz to 100 GHz (Task 4.2), on linear axes and as Bode plots.

On linear axes (top of Fig. 6) nearly all of the decade-wide frequency range is squeezed against the axis. f_β and f_T cannot be read there, and the $1/f$ behaviour looks like a sudden drop. The Bode plot turns a single-pole roll-off into a straight line of -20 dB/decade. Then f_T is the 0 dB crossing, f_β is the point 3 dB below $20\log\beta_0 = 41.6$ dB, and many decades are visible at once. That is why frequency responses are drawn as Bode plots.

11 CS and CE amplifiers (Task 5)

For the CS stage,

$$A_v(s) = \frac{A_M}{1 + s/\omega_H}, \qquad A_M = -\frac{R_G}{R_G + R_{sig}}\, g_m R'_L, \qquad R'_L = r_o \parallel R_D \parallel R_L,$$
$$\omega_H = \frac{1}{(R_{sig} \parallel R_G)\,[C_{gs} + C_{gd}(1 + g_m R'_L)]}.$$

Table 4. Amplifier results (Task 5.2).

	CS (MOSFET)	CE (BJT, $r_x = 100\,\Omega$)
R'_L	4.44 kΩ	2.34 kΩ
A_M	−17.7 (24.9 dB)	−25.5 (28.1 dB)
C_{in}	180 fF	28.7 pF
R'_{sig}	49.8 kΩ	1.36 kΩ
f_H	17.8 MHz	4.08 MHz
GBW = $\lvert A_M \rvert f_H$	314 MHz	104 MHz
0 dB crossing (plot)	314 MHz	104 MHz

The Miller term dominates both input capacitances: $C_{gd}(1 + g_m R'_L) = 120$ fF against $C_{gs} = 60$ fF, and $C_\mu(1 + g_m R'_L) = 23.5$ pF against $C_\pi = 5.2$ pF.

The CE formulas in the assignment apply the input divider to $r_x + r_\pi$. Only the part across r_π drives g_m, so the exact mid-band gain contains a further factor $r_\pi/(r_x + r_\pi) = 0.94$: $A_M = -23.9$ (27.6 dB). Likewise, the resistance seen by C_{in} is $r_\pi \parallel (r_x + R_{sig} \parallel R_B) = 1.29\,\text{k}\Omega$, so $f_H = 4.30$ MHz. The table follows the assignment's formulas, as the task requires.

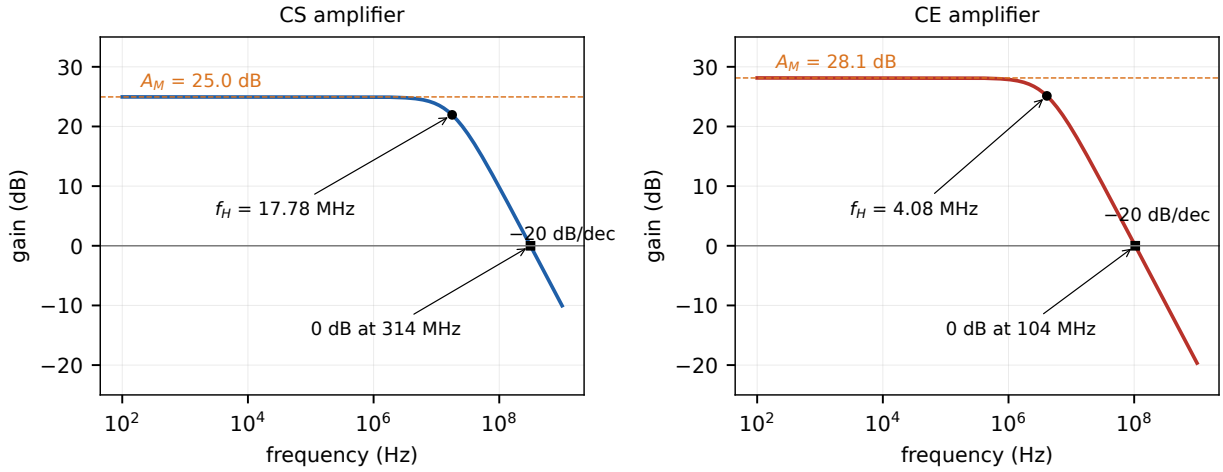


Figure 7. Bode plots of the two amplifiers from 100 Hz to 1 GHz (Task 5.3).

In both plots (Fig. 7) the gain is 3 dB below mid-band at f_H and then falls at −20 dB/decade. Hence the 0 dB crossing equals $|A_M| f_H$. For a single-pole response the gain-bandwidth product is constant, and it equals the unity-gain frequency.

12 RF input matching (Task 6)

With inductive degeneration, $Z_{in} = sL_g + g_m L_s/C_{gs} + 1/(sC_{gs})$.

- The real part is $g_m L_s/C_{gs} = 50\,\Omega$, which gives $L_s = 50\,C_{gs}/g_m$. Since $g_m/C_{gs} \approx \omega_T$, this is $L_s \approx 50/\omega_T$; for the MOSFET above, $L_s = 0.75$ nH.
- Cancelling the imaginary part at ω_0, $\omega_0 L_g - 1/(\omega_0 C_{gs}) = 0$, gives $L_g = 1/(\omega_0^2 C_{gs})$.

With an inductive load, $A_v = -g_m Z_L$ becomes complex and changes rapidly near the tank resonance. The component that links output to input is C_{gd}, whose Miller admittance is $sC_{gd}(1 - A_v)$.

- Below resonance the load is inductive, A_v has a positive imaginary part, and the Miller term contributes a *negative* real part to the input admittance. Above resonance it adds capacitance.
- The equivalent capacitance $C_{eq} = C_{gd}(1 - A_v)$ is therefore complex and can have a negative real part.
- A negative input conductance can cancel the source conductance, which can make the stage oscillate. This is why tuned stages use cascodes or neutralisation to remove the C_{gd} feedback.

13 Tuned amplifier with a transformer load (Task 7)

The load is the parallel combination $Z = R_{total} \parallel sL_p \parallel 1/(sC_{out})$, so

$$A_v(s) = -g_m \frac{s/C_{out}}{s^2 + s/(R_{total}C_{out}) + 1/(L_p C_{out})}.$$

- This is a band-pass response: it has a zero at DC, where the inductor shorts the output, and two zeros at infinity, where the capacitor does.
- The gain peaks at $\omega_0 = 1/\sqrt{L_p C_{out}}$. There the s^2 and constant terms cancel, which leaves the real value $A_v(j\omega_0) = -g_m R_{total}$.
- With $Q = R_{total}\,\omega_0 C_{out}$, the bandwidth is $\text{BW} = \omega_0/Q = 1/(R_{total}C_{out})$, independent of L_p.

The peak gain is proportional to R_{total} and the bandwidth inversely, so their product g_m/C_{out} is fixed. A higher gain can only be bought with a narrower band.

Files

- `part1-state-variable-filter/`: `state_variable_filter.asc` and `saf_notch.asc` (rebuilt).
- `part2-allpass-noise-cancellation/ltspice/`: gain, summer/inverter and delay stages, as built and corrected.
- `part2-allpass-noise-cancellation/simulink/`: `anc_system.slx` (as submitted) and `anc_system_corrected.slx`.
- `verification/`: ngspice netlists, `anc_check.py`, and the scripts for the figures of this report.